

How to: Rebuild the CT-26, and MAKE IT LIVE

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Well after going through two of these on my own, I finally got it right! Its a long post, and hopefully I didn't forget any thing.

So here's my writeup on how to tear down, inspect, and rebuild your CT-26 even if its upgraded.

DISCLAIMER: YOU TRY IT, YOU'RE RESPONSIBLE. NOT ME. DON'T BLAME ME FOR ANY THING IN THIS POST.

First and foremost, remove the waste gate bolts, its "C" clip, all the water lines, oil lines, and the water block off plate that the dipstick attaches to.

Next, remove the clamp that attaches the exhaust housing to the center cartridge.

Typically, the exhaust housing won't come off just by pulling. If it does, you're turbo is a fresh rebuild, or you're just plain lucky! If it doesn't..... Secure the turbo in a vice via the compressor housing that you have yet to take off. **DO NOT CRUSH THE ALUMINUM NOSE! CLAMP IT AROUND THE HOUSING! NO GORRILAS!!!! DO NOT GORRILA IT DOWN INTO SUBMISSION!!!!** Gently heat the area that was under the clamp (two lips). Then, try removing the exhaust housing while wearing a set of welding gloves. If this fails, try using a deadblow non metallic hammer. Lastly, if all else fails.... while the area is still hot from the torch of your choice, take a very fine nosed chisel or cheap screw driver, and chisel the two apart. **BE CAREFUL NOT TO "PRY" IN ONE SPOT! YOU WILL BREAK THE LIPS OFF THE CENTER SECTION!**

After you get the two pieces separated, inspect both for cracks from use and from your efforts of getting them to separate.

Now on to the compressor housing.

Take the biggest nastiest pair of snap ring pliers that you can find, and grab a buddy with two flat head screw drivers. Again, the turbo should still be in the vice as stated above. If the snap ring holes are not where you can get to them, you will need to hammer them with a chisel over to any desired location that you can reach them. Also, **DO NOT WORRY** about the ring coming off and hitting you. IT can't go any where because of the bearing housing it wraps around. Grab the snap ring with the pliers, and chances are it won't come out of its groove. Ask the buddy to insert one of the screw drivers between the housing and outer edge of the snap ring. He/she should be able to pull back on the screw driver and get under the snap ring, forcing it out and away from the housing. Do this all the way round until you hear "PING!", or something similar to a "Ringer" in horse shoes. The snap ring will now be loose around the bearing housing.

Grab the center cartridge (bearing housing) and pull it off the compressor housing.

Release the compressor housing out of the vice. Now, get a socket out of your tool box. It will be a 12 point socket, and IIRC a 17mm. This will fit the turbine or exhaust end. Clamp the socket into the vice. Stand the turbo assy on the socket, with the turbine in the socket. IIRC, the nut on the compressor wheel is a 8mm 12 point. IT IS LEFT HANDED THREAD!!!!

SCRIBE OR DREMEL A LINE ON THE END OF THE TURBINE NOSE, ACROSS THE NUT, AND ONTO THE COMPRESSOR WHEEL NOSE. These are alignment marks that must be made before you continue!!!!!!

Take the nut off. Typically, there will be loc-tite on the nut and in the center of the compressor wheel where the threads are. GENTLY take a rag, and by hand twist the compressor wheel off. If it refuses, hold the center section in your hands vertical over a nice pair of your buddy's hands or a wad of rags and GENTLY tap with a soft face hammer the nose of the turbine. You NEED TO BE EXTRA GENTLE here due to the fact you don't want to A.) damage the threads, B.) bend the nose of the turbine shaft. Chances are that you will actually knock the turbine shaft out the back of the bearing housing. The heat shield will come with it. This is where your buddy's soft hands or lump of rags come in handy: they catch that turbine from dinging or bending the fins when it drops.

Set the compressor wheel and nasty turbine aside, and go back to the area under where the compressor wheel used to be. Take your snap ring pliers again, and pull the seal plate off. Stuck inside the seal plate, will be your front seal. It is wrapped around the front seal washer. Further down, you will remove the thrust bearing, and the thrust washer. After that, you will find a brass bearing, a center steel bearing spacer, and then another brass bearing.

Remove the two rubber seals in the compressor end of the bearing housing: one under the compressor housing (may still be in the compressor housing) and the other in the bearing housing where the front seal plate was. Discard these IF YOU HAVE new ones in your rebuild kit (which you should or your kit sucks).

Now, onto inspection and cleaning.

You can sand blast

- The rear seal area of the turbine (VERY LIGHTLY! Do not blast in the same place for long periods of time)
- The rear heat shield (very thin, be careful)
- The exhaust housing. You can blast on this until your little heart is content.
- The compressor housing. WORD OF CAUTION. Do not blast aluminum and cast iron at the same time in the same cabinet. Also, do not sand blast the inlet of the turbo, or the seal area on the back of the housing. Again, DO NOT blast in one area for long periods of time.

As for the bearing housing, turbine fins, and compressor wheel, you can soak them in any solvent of your choice as long as its not a harsh alkali or acid. Acetone works well, as does carburetor dip from napa or any mineral spirit solution / brake cleaner. Also, some abrillo or scotchbrite pad won't hurt to be used on the bearing housing.

Make sure all the coked oil in the rear area of the bearing housing is removed! A small stainless steel brush and brake cleaner does wonders here.

Inspect the rear seal area on the turbine (there are two grooves, one round, one square. The one closest to the turbine fins is the seal area). Make sure the groove is still square, not rounded, chipped, or offset grooved once cleaned. If it is, its time for a new turbine shaft.

Inspect the turbine and compressor wheels for chipped, bent, or odd (missing) fins. Again, this is catastrophic and the part is not salvageable. Replace it.

Machine shop work.

Take the bearing housing once clean to a reputable reliable machine shop. Ask them to check the bore of the housing. If its remotely straight, ask them to REAM (not bore or machine) the inside of the housing to 0.6350 in. Make SURE THEY ONLY BORE THE BEARING AREA INSIDE THE HOUSING. They should not have to come in contact with the rear seal lip. You may have to return to the machine shop again, and I will tell you more about this later.

Assembly.

You should have your rebuild kit. Order a rebuild kit with bearings that have an OD (outer diameter) of 0.6315 in. (should be 0.0005 in. over). The bearings should measure IIRC 0.3530 in. in height. The bearing stack (both bearings and steel spacer) should measure IIRC 1.9235 in.

Take the bearings out and let them soak in oil over night. This should be the same oil that you are using in your car.

Take the front seal washer, and the thrust washer out of the package. The thrust washer should measure 0.2935 in. in thickness. Both of the washers put together are called the thrust stack. The thrust stack should measure 0.6685 in.

Take or send the turbine shaft, the NEW thrust stack, the compressor wheel, and the new nut to a turbo or diesel supply company that can balance it. Remember those marks you make on the nose of the turbine, nut, and compressor wheel nose earlier? You're turbo balance job now depends on this to be right.

When your turbine assy returns, take a magic marker and mark a line on the sides of the stack (thrust washer and front seal washer) directly in line with the alignment marks you made earlier on the nose of the turbine. The assy should be tight together with the nut at the end. Wait to disassemble this.

Take your bearing cartridge (center section) and again make sure its clean and free of all dirt, debris, and any gasket or silicone that might be left over on any surface.

Take your bearings out of their oil bath. Make sure they freely pass through the center

section, with NO PLAY. If they do not pass, take the center section back to the machine shop. Increase the bore 0.0005 in. Set the bearings aside with the steel spacer.

Disassemble the balanced turbine assy. Make sure it was balanced with YOUR alignment marks and not another set.

Take your turbine shaft, again free of any debris and deposits, and install the rear piston ring seal. Be very careful not to bend the ring seal or break it. Gently oil it. Position the seal onto the shaft so that the gap is 180 deg from the oil exit of the center section. The gap should be up. Install the rear heat shield onto the center section. Gently press the shaft with your hands into the center section. You will hear a light snap of the seal into the housing. The shaft should be able to spin. Take your bearings, and generously lube them with oil. Install the rear bearing, then the steel spacer, and last the other bearing.

This next step is controversial, but it is for good reason! The thrust bearings from most rebuild kit companies are NOT FLAT. It needs to be 100% flat in order to work 100% of the time 100% correctly!

Take out the thrust bearing from the kit, and take a 700-1000 grit piece of oiled sandpaper. Put the sand paper on a piece of flat glass, or a clean smooth FLAT surface. In a circular motion, sand each side of the bearing. With this fine of sand paper, all you are doing is making sure the bearing is flat. Each side should be nice and shiny (almost chrome like). After this, make sure you clean it VERY WELL with brake cleaner, and compressed air. Make sure the little oil holes are clean and clear of any dirt from the sand paper.

After cleaning, oil it. Install the Thrust washer around the thrust bearing, flat side up. Make sure to align the magic marker line on the thrust washer in line with the scribed line on the nose of the turbine. Install the thrust washer and bearing with the notch around the oil return hole, and the alignment pin in the hole in the bearing. (It will only install in one direction if the housing has an alignment pin which most do).

Oil the small rubber o ring seal that came in the kit. This is the front seal plate o ring. Install it into the groove in the center section.

Note: On some early or possibly late (not sure when Toyota or Hitachi made the change) there will be a metal baffle pressed into the back of the front seal plate. This baffle is NOT NEEDED. If you choose not to install it you have nothing to worry about.

Now, take the front seal washer, and install the NEW front piston ring seal around it. You will need to mark with your magic marker a line up, around, and on top of the front seal washer so that you can align it with the scribe mark on the turbine shaft nose. Again there are two grooves here, install it on the one farthest forward on the nose of the front seal washer. Oil the front piston ring seal. Very carefully install the front seal washer with piston ring around it, into the front seal plate. Be sure the piston ring seal

doesn't try to slide sideways or down to the next groove. Oil the outside of the front seal plate. Again, install the gap of the piston ring seal up, or 180 deg away from the oil return in the center section. Install the front seal plate assy into the bearing housing, with the front seal washer alignment mark in line with the line on the nose of the turbine shaft. Take the snap ring you removed earlier from this location, and FLAT SIDE DOWN, TAPERED side up.

Install the compressor wheel with its alignment mark in line with the mark on the nose of the turbine shaft.

Take the whole assy that you have just built, and let it sit upright again in the socket that should still be in the vice.

Install the compressor wheel nut. Again, the nut is LEFT HAND THREAD.

Torque on the nut is 8-10 foot lbs IIRC. Some rebuilders recommend A SMALL dab of BLUE loc-tite here on the threads. This to me though is redundant. The threads are left hand, and the nut will try to tighten itself under normal operating conditions. This all being said that your turbo doesn't get subjected to horrible surge problems or a bad BOV.

Take the now finished center section out of the socket. You should be able to spin VERY FREELY the compressor wheel and turbine. If you can not, you have an incorrect thrust stack or bearing stack inside. Go back and double check to make sure you installed the thrust washer on the bearing correctly, flat side up.

If the assy does indeed spin freely, clamp the compressor housing back into the vice. Install the large rubber o ring from the kit into the compressor housing. Make sure its oiled. Put the large snap ring around the center section, flat side up against the center section back plate. Align the compressor housing hole or notch, with the pin or notch in center section. Install the snap ring.

Take some anti seize and lightly put it around the inside of the rear housing where it attaches to the center section. Install the clamp around the center section and rear housing. Do not TIGHTEN the nuts at this time. Snug them only. You will need to reposition the clamp once you get the turbo on the car so you can get to the nuts on the mounting flange.

Install the waste gate actuator, the waste gate actuator c clip, the water lines, and the oil lines back on the turbo. Make sure you have a good gasket or seal that SEPARATES the oil feed and oil return on the turbo. They are on the same flange. BAD DESIGN! If the two are not separated, your turbo will have no oil pressure and will die! When you have every thing installed, put your finger over one side of the oil feed banjo fitting. Fill the oil feed line with oil. Spin the compressor wheel with your hand for about 30 seconds. This ensures that the bearings inside see oil at startup. Don't worry about the lack of oil coming back out at this time. It will all come out the return tube when you install the turbo.

Install the turbo. Pull your EFI fuse. Crank the car over several times to ensure the turbo is primed and has oil in the line. Reinstall the EFI fuse, and start the car. Let the car run and warm up several minutes.

Turn the car off. Inspect the compressor wheel for play. If it has play, rebuild time. The turbo is already dead.

If no play is evident, put all intake parts back together, drive car. ENJOY!